

MESSOU FRANCK

AI Research Engineer · LLMs & Time-Series · Graph Neural Networks · Edge ML for IoT & Energy

@ franckaboya99@gmail.com

📍 Tokyo, Japan

in [mesabo](#)

🔗 [mesabo](#)

🔗 [mesabo.github.io](#)

🎓 [Google Scholar](#)

PhD candidate and ML engineer with 3+ years advancing LLM / small-language-model reprogramming for time-series forecasting and imputation, graph neural networks for power-system stability, and edge / IoT sensor intelligence. First-author of **2 Q1 journal papers** and **5 IEEE conference papers** (GLOBECOM, ICC, VTC, ICUFN), with the IEEE VTS 2024 Student Paper Award. Bridges research and production: LoRA / PEFT fine-tuning, distributed training (DDP), and vLLM serving on multi-GPU infrastructure, plus LangChain / LangGraph / RAG pipelines, alongside shipped AI products (CVFlow, AIReco). Available immediately for a **research engineer or applied-scientist role** in AI / ML.

KEY ACHIEVEMENTS

- First-author of **2 Q1 journals** (Internet of Things IF 7.6; Expert Systems with Applications IF 7.5) and **5 IEEE conference papers**; IEEE VTS 2024 Student Paper Award.
- Pioneered LLM / sLM reprogramming (Spec2Llama, Spec2LLM, LLM4Imp) for forecasting and imputation on edge, published at **IEEE ICCE / ICC 2026**.
- Designed dual-domain graph-neural-network architectures for power-system stability on the PEGASE **2,869-bus** grid.
- Provisioned and operate a multi-GPU server end-to-end (DDP, checkpointing, vLLM) shared across the team.

EXPERIENCE

AI / ML Researcher (Doctoral)

YuLab, Hosei University

📅 Sep 2025 – Present

📍 Tokyo, Japan

- Built LLM / sLM reprogramming pipelines (Spec2Llama, Spec2LLM, LLM4Imp) adapting frozen LLMs and compact backbones (SmolLM, Phi, Qwen) for forecasting and imputation on edge, updating **under 9% of parameters** while matching or beating SAITS / TimeMixer baselines; published at IEEE ICCE / ICC 2026.
- Designed graph-constrained multi-LLM ensembles with structural causal models, lifting causal-discovery **F1 to 0.945** (+11 pts over PCMCi) and narrative completeness **5.6×** over a single-LLM baseline.

ML Researcher (Master's)

YuLab, Hosei University

📅 2023 – 2025

📍 Tokyo, Japan

- Built transformer time-series forecasting models (TAPformer, TSFormer, CNN-BiLSTM) for long-term electricity load forecasting, outperforming standard CNN / LSTM baselines (published in *Internet of Things*, Q1, IF 7.6).
- Won the IEEE VTS Tokyo / Japan 2024 Student Paper Award (VTC2024-Fall, Washington D.C.).

LLM Systems Engineer

Open-Source / Personal Research Initiative

📅 2025 – Present

📍 Remote · [GitHub.com](#)

- Architected and deployed four open-source production LLM systems (semantic search, retrieval-augmented generation, stateful agent orchestration, full-stack LLMOps), shipped via **GitHub Actions CI/CD** to GCP (Cloud Run, GKE, Vertex AI).

Text in blue is clickable — it opens the linked profile, project, or paper.

EDUCATION

PhD, Applied Informatics

Hosei University

📅 2025 – 2027 (Expected)

📍 Tokyo

MSc, Applied Informatics

Hosei University

📅 2023 – 2025

📍 Tokyo

TECHNICAL SKILLS

Python PyTorch Hugging Face

Transformers LLM Fine-tuning

LangChain LangGraph LangSmith

RAG RAGAS FAISS Vertex AI

LoRA / PEFT Time-Series Forecasting

Imputation GNN Anomaly Detection

vLLM CUDA DDP Docker FastAPI

scikit-learn

SOFT SKILLS

Research Mindset Scientific Writing

Cross-cultural Teamwork Communication

Adaptable Collaborative

APPLIED WORK

Co-Founder & Lead

CVFlow ([cvflow.app](#))

📅 2025 – Present

📍 Remote

- Co-built a production AI resume platform; owned the LLM integration (Gemini / GPT) across 20 languages.

PARTICIPATION

- Engineered composable LCEL pipelines with hybrid retrieval (FAISS, ChromaDB, BM25 sparse, MMR re-ranking), evaluated with RAGAS (context recall, faithfulness, semantic similarity).
- Built stateful LangGraph agents (conditional routing, cycles, human-in-the-loop) and ReAct loops with Pydantic-validated tools; FastAPI streaming APIs with semantic caching and LangSmith tracing at sub-200ms TTFT.

LangChain LangGraph LangSmith FAISS ChromaDB
RAGAS GCP

Machine Learning Engineer
D'isum Inc.

📅 Oct 2025 – Jul 2026 📍 Tokyo, Japan — Hybrid

- Built ETL pipelines from raw sensor streams (50+ engines, 200+ batteries) and classical ML models for sound and time-series anomaly / fault detection.
- Built on-device iOS / Android sensor apps (Swift, Java) for real-time data capture and edge inference.

Generative AI Intern (Summer Internship)
日本AIコンサルティング (Japan AI Consulting)

📅 Summer 2024 📍 Osaka, Japan

- Customized AI model source code and integrated LLMs into interactive UIs using Gradio, delivering tailored outputs for internal use cases.
- Developed and maintained log data processing pipelines using FastAPI and Flask, documented via Growi.
- Produced research reports on data privacy and the AI industry landscape, supporting J-AIC's business expansion strategy.

RESEARCH PROJECTS

SmolSearch – Semantic Search API

FAISS · LCEL · FastAPI · Cloud Run · [Github.com](#)

- ★ LCEL chain (SmolLM2-360M) over a FAISS index; FastAPI on Cloud Run, sub-100ms TTFT, MMR search.

RAGify – Evaluated RAG System

ChromaDB · RAGAS · GCP · [Github.com](#)

- ★ Hybrid retrieval (ChromaDB, FAISS, MultiQuery, BM25) scored with RAGAS on GCP.

AgentFlow – Stateful Multi-Tool Agent

LangGraph · ReAct · LangSmith · GKE · [Github.com](#)

- ★ LangGraph StateGraph with Pydantic tools and full LangSmith tracing; deployed on GKE.

LLMOps Baseline – Production LLMOps Reference

LangChain · LangSmith · Vertex AI · FastAPI · [Github.com](#)

- ★ LCEL + Pydantic output, Redis caching, SSE streaming, tiktoken costs; GitHub Actions CI/CD to Vertex AI.

ACADEMIA

Research Assistant (part-time)
YuLab, Hosei University

📅 2023 – Present 📍 Tokyo, Japan

- Assist faculty with data curation and paper drafting; TA for English-medium undergraduate and Master's classes.

- Presenter at IEEE international conferences (VTC, ICUFN, GLOBECOM, ICC); represented J-AIC at JICA's TICAD9 booth, 2025.

AWARDS & CERTIFICATES

- IEEE VTS Tokyo / Japan Chapter 2024 Student Paper Award & Encouragement Award (VTC2024-Fall).
- JICA Development Studies Program (JICA専用講座), 2025.
- Review Committee Member, IEEE ICECET 2026 (IEEE Italy Section); Reviewer, Theoretical Computer Science (Elsevier).

SELECTED PUBLICATIONS

- TAPformer: Long-Term Electricity Load Forecasting in IoT Systems, *Internet of Things* (Elsevier), 2026 — Q1, IF 7.6.
- Spec2Llama: Spectral-Aware Forecasting for Versatile Time Series, *Expert Systems with Applications*, 2026 — Q1, IF 7.5.
- Hybrid Attention CNN-BiLSTM for Short-Term Load Forecasting, *IEEE VTC2024-Fall* — Student Paper Award.
- TSFormer: Temporal-Aware Transformer for Multi-Horizon Forecasting, *IEEE ICUFN* 2025.
- Spec2LLM / LLM4Imp: LLM reprogramming for forecasting and imputation, *IEEE ICCE / ICC* 2026.

LANGUAGES

French (native) ●●●●●

English (fluent) ●●●●●

Japanese (daily, ongoing) ●●●●●